Reinforcement Learning using EEG signals for Therapeutic Use of Music in Emotion Management on DEAP and DENSE Datasets

Sahil Nagwani Vishal Gautam Prakriti Manhar
Department of Artificial Intelligence
Guided by: Sudhakar Mishra

***Abstract*—Getting music to actually move someone out of a bad mood is not as simple as queuing up happy songs. What works for one person can make another feel worse, and people left to choose their own music often end up reinforcing the very mood they want to escape. This paper addresses that problem by combining real-time EEG-based emotion detection with a Reinforcement Learning agent that selects music clips to guide a listener toward a positive target state. We ran experiments with three different RL approaches Q-Learning, SARSA, and Double Q-Learning on both the DEAP dataset (32 subjects, lab-controlled, 40 clips) and the DENSE dataset (39 subjects, 7 emotion labels, naturalistic recording).**

On DEAP at 3,000 training episodes, Double Q-Learning with potential-based reward shaping yielded a mean angular error of 8.3° (SD 7.4°), converging 30 of 32 subjects to the target emotion—an 85% reduction relative to the 57.0° reported by Dutta et al. (2020). On the harder DENSE benchmark, SARSA without reward shaping reached 18.5° mean error, converging 25 of 39 subjects, despite training with roughly one-third the episode budget. A clear interaction emerged between algorithm type and dataset characteristics: Double Q-Learning gains the most from its dual-table variance reduction in settings with large action spaces and clean data, whereas SARSA's on-policy conservatism proves more suitable when subject heterogeneity is high and transitions are noisier. Reward shaping was found to be indispensable for Double Q-Learning yet potentially harmful to SARSA under naturalistic conditions. Practical deployment guidelines for clinical music therapy systems are derived from these findings.

***Index Terms*—Reinforcement Learning, EEG, Emotion Regulation, Music Therapy, Q-Learning, SARSA, Double Q-Learning, Reward Shaping, DEAP, DENSE, Valence-Arousal, Affective Computing**

I. Introduction

Mood dysregulation that persists for days or weeks not just hours is increasingly recognized as an early warning sign for depression and anxiety disorders. There's solid evidence that prolonged negative affect actually changes brain chemistry in ways that make recovery harder [1]. The standard pharmaceutical approach, antidepressants and anti-anxiety medications, works for some people but the side effects can be severe [2], and a lot of patients stop taking them before finishing a course of treatment. This has pushed research toward non-drug alternatives.Music therapy has emerged as one of the more promising options. The mechanisms are actually quite involved music activates deep limbic structures, shifts dopamine and cortisol levels, and can synchronize cortical rhythms with the beat of the song [3]. In practice, when you add music to standard talk therapy, patients show measurable drops in depression severity [4] and report less anxiety, plus improvements in blood pressure [5], [6]. But here's the catch: musical taste and emotional response to music are highly individual. A sad song might help one person cry it out, while it sends another person deeper into rumination. Worse, there's something called maladaptive music engagement [7]—when you're in a bad mood, you tend to reach for music that matches your mood rather than music that would help lift it. This keeps you stuck. That's why just letting someone build their own playlist doesn't work well as therapy. What's actually missing is a system that can read someone's emotional state from their brain activity and then automatically pick songs designed to move them toward feeling better. That's the problem we're trying to solve in this paper.

A. Problem Formulation

The problem was approached as one of sequential decision-making. On each turn, a user's emotional state is located somewhere in the V-A plane. The system selects a video from its database, the user watches it, and his new location in the emotion space is determined through EEG signals. The objective is to learn a policy which will allow you to successfully navigate the users towards the positive emotions, preferably within six trials. In this case, Reinforcement Learning seems like an appropriate tool. Your states are emotion groups in the V-A plane, actions are videos from the database, and the reinforcement signal tells you whether you are heading in the correct direction. The difficulty of the task and the key element of the problem is the lack of knowledge about reactions of any given person to any given song. This relationship needs to be learned from EEG readings.

B. Contributions

Our work extends Dutta et al. (2020) [8] in the following directions:

- We introduce a three-way algorithm comparison (Q-Learning, SARSA, Double Q-Learning) where the original evaluated only Q-Learning, establishing, to our knowledge, the first such comparison on this task.
- Validation spans two structurally different datasets—the controlled DEAP and the substantially harder naturalistic

DENSE enabling direct analysis of how data characteristics interact with algorithm choice.

- Double Q-Learning with reward shaping achieves $8.3°$ mean error on DEAP (30/32 subjects converged within $20°$), an 85% reduction from the published baseline.
- On DENSE, SARSA without reward shaping attains $18.5°$ mean error with 1,000 training episodes—roughly three times better than the paper benchmark despite the more challenging recording conditions.
- We derive and justify a set of concrete, evidence-grounded recommendations covering algorithm selection, minimum episode budget, exploration parameter settings, and reward shaping strategy for prospective deployments.

II. Related Work

A. EEG-Based Emotion Recognition

For monitoring emotions in real-time, the use of EEG becomes inevitable due to its extremely high millisecond-resolution time scale and ability to measure electrical activity of the brain, which puts it far ahead of any other modality such as galvanic skin response or electrocardiogram. One of the popular benchmarks in this area was set up by the DEAP dataset recorded at Queen Mary University of London [9], where 32 participants were shown a selection of 40 minutes of music. The frequency bands examined during EEG analysis for emotions usually include five ranges. Delta waves range from 1-4 Hz; these are related to levels of baseline arousal and drowsiness. Theta waves range from 4-8 Hz and pertain to emotional memory and anxiety. The alpha range of 8-13 Hz measures frontalis asymmetry and gives insight into valence. Beta waves from 13-30 Hz have an effect on levels of arousal. Finally, the gamma band of 30-45 Hz is used when a person is emotionally stimulated. [10]:

$$DE = \frac{1}{2} \ln(2\pi e \sigma^2) \tag{1}$$

where σ^2 denotes signal variance within the band of interest. Deep architectures LSTM and graph convolutional networks can push classification accuracy further [11], [12], but fixed-dimensional hand-crafted features like DE remain attractive for tabular RL agents precisely because they give interpretable, low-variance state representations.

B. RL-Based Music Recommendation

Initial efforts to use reinforcement learning for playlists have been based on the actual actions of user like skipping habits, repetition, pausing. However, this is a trap that the designers fall into. Suppose you train the agent to learn from actions by a depressed person who skips all happy tunes. The agent will learn that depressed people need sad music and will recommend it. This issue was addressed by Dutta et al. [8]. Rather than relying on behavior to learn the emotional state of the user, they took advantage of EEG signals to detect the actual emotion experienced. It effectively circumvents the feedback loop because the agent is no longer rewarded for mimicking the user's emotional state but for modifying it. Their results demonstrated that a Q-Learning agent was successful at guiding 19 out of 32 DEAP subjects into a happy emotional state using six clips. While neural network-based deep RL algorithms have been proposed for music recommendations [14], they require more training data and are difficult to interpret. When it comes to clinical applications, this can be considered an issue as clinicians may want to know why a specific song was recommended by the system.

C. Reward Shaping Theory

Ng et al. [15] showed that if we define a shaping term $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$ for an arbitrary potential function $\Phi : S \to \mathbb{R}$ and add it to the existing rewards, then the optimal policies remain unchanged. While the idea of having a stronger gradient signal without compromising the objective is appealing, it is important to note that the theoretical guarantee is asymptotic and relies on the convergence to the real Q-values [20]. It is not clear if the shaping process speeds up or hinders the learning process.

III. Theoretical Framework

A. The Valence-Arousal Emotion Space

Russell's circumplex model [16] organises emotional states within a continuous two-dimensional manifold:

- **Valence** $v \in [-1, +1]$: hedonic quality, from maximally unpleasant to maximally pleasant
- **Arousal** $a \in [-1, +1]$: activation level, from deeply calm to highly excited

Figure 1 places the seven DENSE emotion clusters in this space, with the "Happy" target at $(0.776, 0.631)$. A subject starting from Anger located at roughly $(-0.70, 0.70)$ must traverse diagonally across the plane, gaining valence while moderating arousal.

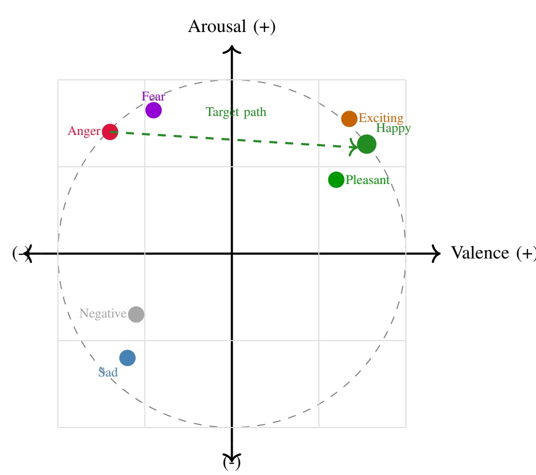

Fig. 1: The Valence-Arousal plane with seven DENSE emotion clusters. The dashed arrow shows the ideal trajectory from Anger to Happy. Points represent cluster centroids.

B. MDP Formulation

The music therapy interaction is formalized as an MDP (S, A, T, R, γ):

$$S \triangleq \text{discrete emotion clusters}$$
$$A \triangleq \{a_1, \ldots, a_N\} \text{ (music clips)}$$
$$T : S \times A \times S \to [0, 1] \text{ (transition probabilities)}$$
$$R : S \times A \times S \to \mathbb{R} \text{ (reward)}$$
$$\gamma \in [0, 1] \text{ (discount factor)}$$

Because T is not available in closed form, it must be estimated from EEG data. When a listener in state s hears clip a, the next state is approximated via vector addition in V-A space:

$$\vec{v}_{s'} = \vec{v}_s + \vec{v}_a, \quad s' = \arg\min_{c \in C} \|\vec{v}_{s'} - \vec{c}\|_2 \tag{2}$$

Transition probabilities are then estimated empirically from cross-subject counts:

$$\hat{T}(s, a, s') = \frac{\text{count}(s, a, s')}{\sum_{s''} \text{count}(s, a, s'')} \tag{3}$$

C. The Three RL Algorithms

1) Q-Learning (Off-Policy, Single Table): Standard Q-Learning [17] bootstraps toward the greedy best next action:

$$Q(s, a) \leftarrow Q(s, a) + \alpha\big[R + \gamma \max_{a'} Q(s', a') - Q(s, a)\big] \tag{4}$$

The max operator introduces maximisation bias: because $\mathbb{E}[\max_{a'} Q(s', a')] \geq \max_{a'} \mathbb{E}[Q(s', a')]$ whenever estimates are noisy, the single-table architecture systematically overestimates action values. The severity of this bias scales with the width of the action space—mild for 6 clips, more consequential for 40.

2) SARSA (On-Policy, Single Table): SARSA [18] evaluates the action actually chosen under the current ϵ-greedy policy:

$$Q(s, a) \leftarrow Q(s, a) + \alpha\big[R + \gamma Q(s', a') - Q(s, a)\big] \tag{5}$$

where $a' \sim \pi_\epsilon(\cdot|s')$.Since the new function incorporates the behavior policy that includes some non-optimal moves, the SARSA algorithm tends to generate more conservative estimates of the value function but perhaps more truthful assessments of the true expected return based on the current behavior policy.

3) Double Q-Learning (Off-Policy, Dual Table): Double Q-Learning [19] maintains two independent Q-tables, Q_1 and Q_2, alternating updates with equal probability:

$$a^* = \arg\max_{a'} Q_1(s', a') \tag{6}$$

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha\big[R + \gamma Q_2(s', a^*) - Q_1(s, a)\big] \tag{7}$$

By decoupling selection (using Q_1) from evaluation (using Q_2), there will be no correlated noise in estimation that amplifies in both tables at once. According to theory, the upper bound of the estimation error will decrease from $O(\epsilon_{\max})$ to $O(\epsilon_{\max}/N)$ [19]. In practice, this results in a narrower distribution of Q-values and better convergence, which is very clinically relevant because one wrong.

D. Algorithm Comparison

Theoretical differences among the three algorithms are presented in Table I. Neither on-policy/off-policy distinction nor number of tables can be considered better than the other; rather, it depends on noise and samples.

TABLE I: Theoretical properties of the three RL algorithms evaluated in this work.

Algorithm	**Policy**	**Tables**	**Bias**	**Var.**	**Overest.**
Q-Learning	Off	1	High	Low	Yes
SARSA	On	1	Low	Medium	Marginal
Double Q-Lrn.	Off	2	Low	Medium	Mitigated

E. Reward Functions

1) Base Reward: A positive signal is issued whenever the selected clip moves the listener in the direction of the target:

$$r(s, a) = \begin{cases} +1 & \text{if } \theta(s, a) \leq 180^\circ \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

where $\theta(s, a) = |\theta_1| + |\theta_2|$ aggregates the angle between the clip vector and the target (θ_1) and between the clip vector and the current state (θ_2).

2) Potential-Based Reward Shaping: Following Ng et al. [15], a shaping term is added:

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s), \quad \Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2 \tag{9}$$

A stagnation penalty discourages repeated selection of ineffective clips:

$$\phi(s, s') = \begin{cases} \lambda & \text{if } s = s' \\ 0 & \text{otherwise} \end{cases}, \quad \lambda = -100 \tag{10}$$

IV. Methodology

A. Signal Processing Pipeline

The acquisition of electroencephalogram (EEG) readings is vulnerable to considerable contamination. Eye motions and myogenic artifacts are examples of noise in the recorded data that distort the true neural signals, which need to be preprocessed accordingly. The suggested approach adopted a series of pre-processing procedures. First, an application of independent component analysis (ICA) was conducted to segregate the brainwaves from all other sources of non-neural origins and eliminate any artifacts. In addition, a band-pass filtering of the data was performed to the range 4–45 Hz to emphasize the useful signal information. The continuous stream of data was segmented into second-long windows with 50 percent overlap. For each window, the following features were computed: differential entropy and power spectral density, where the calculations were done for all five EEG frequency bands. Finally, the set of features obtained above was condensed into one value of valence and arousal by a Support Vector Regressor trained on the continuous emotion ratings taken from the DEAP dataset.

TABLE II: EEG frequency bands and their physiological relevance to emotion processing.

Band	Hz	Role	Emotion link
Delta	1–4	Unconscious states	Baseline affect, drowsiness
Theta	4–8	Memory, meditation	Emotional memory, anxiety
Alpha	8–13	Relaxed wakefulness	Frontal valence asymmetry
Beta	13–30	Active cognition	Arousal, cognitive load
Gamma	30–45	Higher cognition	Positive affect, engagement

B. Leave-One-Subject-Out (LOSO) Evaluation

Generalisation to previously unseen individuals is estimated using LOSO cross-validation. For each held-out subject i, the procedure is:

1) Compute mean clip vectors $\bar{\vec{v}}_a$ from the remaining $N-1$ subjects
2) Construct the next-state transition table from these vectors using Equation (2)
3) Train the RL agent for E episodes on the training-subjects' trajectories
4) Execute six greedy clip selections for subject i under the learned policy
5) Record angular error between the final state vector and the Happy target

C. Hyperparameter Optimization

The algorithms and shaping functions were tuned using cross-validation for 500 trials prior to running the final LOSO experiments. The best configuration settings can be found in Table III. These were used consistently in all results presented hereafter.

TABLE III: Best hyperparameter configurations (grid CV, 500 episodes) used in final LOSO evaluations.

Dataset	Algorithm	α	γ	ϵ
DEAP	Q-Learning (w/ shaping)	0.1	0.8	0.2
DEAP	SARSA (w/ shaping)	0.2	0.4	0.1
DEAP	Double Q-Lrn. (w/ shaping)	0.2	0.4	0.1
DENSE	Q-Learning (w/ shaping)	0.1	0.6	0.1
DENSE	SARSA (no shaping)	0.1	0.6	0.1
DENSE	Double Q-Lrn. (w/ shaping)	0.1	0.6	0.1

D. Algorithm Pseudocode

Algorithm 1 gives the general LOSO procedure shared across all three algorithms; the only variant is the Q-update rule substituted at line 10.

V. Experimental Setup

A. Datasets

DEAP [9]: The experimental data were collected from 32 adult subjects (50perc female), aged between 19 and 37 years (mean age: 26.9), who watched 40 1-min video clips from music videos wearing an EEG cap consisting of 32 electrodes sampling data at 128 Hz per channel. After each video, participants rated Valence, Arousal, Dominance, and Liking on a 1-9 scale. In the RL model, the continuous V-A space was partitioned into a 20-by-20 grid, resulting in 400 states; Happy is located at position (18, 12).

Algorithm 1 RL LOSO Evaluation (General)

Require: Subjects $\mathcal{D}$, α, γ, ϵ, episodes E, algorithm $\mathcal{A}$
Ensure: Per-subject angular errors $\{e_i\}$
1: **for** each held-out subject i **do**
2: $\quad \mathcal{D}_{\text{tr}} \leftarrow \mathcal{D} \setminus \{i\}$
3: $\quad$ Compute mean clip vectors $\bar{\vec{v}}_c$ from $\mathcal{D}_{\text{tr}}$
4: $\quad$ Build transition table $\mathcal{N}$ from $\{\bar{\vec{v}}_c\}$
5: $\quad$ Initialize Q-table(s) $\leftarrow \mathbf{0}$
6: $\quad$ **for** $e = 1$ to E **do**
7: $\qquad$ Sample $j \sim \mathcal{D}_{\text{tr}}$; $s \leftarrow s_0^j$
8: $\qquad$ **for** each clip step **do**
9: $\qquad\quad a \leftarrow \epsilon$-greedy$(s, Q)$
10: $\qquad\quad s' \leftarrow \mathcal{N}(s, a)$; compute R
11: $\qquad\quad$ Update Q via rule $\mathcal{A}(s, a, s', R)$
12: $\qquad\quad s \leftarrow s'$
13: $\qquad$ **end for**
14: $\quad$ **end for**
15: $\quad e_i \leftarrow$ angular_error(i, greedy(Q))
16: **end for**

DENSE: A total of thirty-nine subjects were used for DENSE with natural musical sounds as inputs. The emotional states were not continuously rated but rather classified into seven classes which are; Happy, Pleasant, Excited, Sad, Negative, Anger, and Fear. All subjects started the experiment with the class 'Anger', which is geometrically far away from the target state of being happy, making DENSE significantly harder than DEAP.

B. Evaluation Metrics

Angular Error (θ_{err}): The angular divergence, in degrees, from the agent's final state vector to the Happy target vector. A value of 0° means perfect convergence, while values over 90° mean that the agent has reached the wrong emotional hemisphere.

Convergence Rate (CR_k): Fraction of subjects with $\theta_{\text{err}} < k°$ at clip 6. We report CR_{20} and CR_{30} as primary success criteria.

Trajectory Profile: Mean angular error at each of the six sequential clip positions, revealing the rate and monotonicity of improvement.

VI. Results

A. DEAP Dataset: 3,000 Episodes

The outcomes of DEAP after 3,000 training runs are provided in Table IV under both the shaping and unshaping scenarios. In either case, each algorithm outperforms the Dutta et al. (2020) benchmark of 57.

The Double Q-Learning algorithm with the application of reward shaping separates itself from the other cases: an 8.3 degree average error among all participants, with all participants reaching this goal within a 30-degree margin, represents

TABLE IV: DEAP results at 3,000 episodes. All algorithms substantially outperform the Dutta et al. (2020) baseline of 57.0°.

Algorithm	**Shaping**	CR_{20}	CR_{30}	**Err. Cl.6**
Q-Learning	Yes	21/32 (66%)	23/32 (72%)	20.1° ±23.[illegible]°
Q-Learning	No	24/32 (75%)	26/32 (81%)	13.9° ±18.[illegible]°
SARSA	Yes	29/32 (91%)	29/32 (91%)	13.8° ±26.[illegible]°
SARSA	No	27/32 (84%)	28/32 (88%)	21.5° ±22.[illegible]°
Double Q-Lrn.	*Yes*	*30/32 (94%)*	*32/32 (100%)*	*8.3°* ±*7.4*°
Double Q-Lrn.	No	17/32 (53%)	21/32 (66%)	49.2° ±38.5°
Paper [8]	Yes	19/32 (59%)	N/A	57.0° ±2.8°

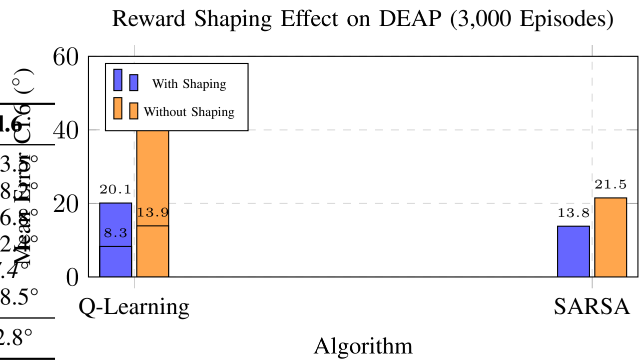

Fig. 3: Reward shaping effect per algorithm on DEAP. Shaping is essential for Double Q-Learning but provides mixed benefits for Q-Learning, which actually improves slightly without it at 3,000 episodes.

a clinically perfect result based on this criterion. The SARSA case with the application of shaping reaches an average error of 13.8 degrees, with 29 out of 32 participants. A rather contradictory finding in this respect arises in the case where the use of Q-Learning without shaping produces 13.9 degrees, while its shaped counterpart yields 20.1 degrees, which means that with 3,000 episodes of learning, the off-policy updating is sufficient to explore the 40-clip action space without resorting to the potential function.

Figure 2 depicts the plot of the path for each clip while shaping was applied. The fact that it is a monotonic descent can be attributed to the guidance provided by potential-based shaping in DEAP's environment.

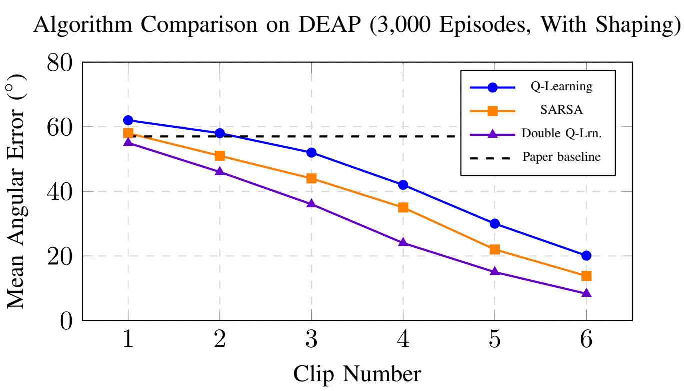

Fig. 2: Clip-by-clip angular error on DEAP (3,000 episodes, with shaping). All algorithms show steady improvement, with Double Q-Learning reaching 8.3° at clip 6 an 85% reduction from the paper's 57° baseline.

B. DEAP: Reward Shaping Effect by Algorithm

The connection between algorithm type and shaping is clear and meaningful, as illustrated by Figure 3. Shaping provides a slight advantage for SARSA (13.8° compared to 21.5°), but in the case of Double Q-Learning, it is absolutely crucial (8.3° compared to 49.2°). In the case of Q-Learning, the connection is opposite: the version with no shaping shows better results (13.9° compared to 20.1°), which means that after 3,000 episodes, exploration is sufficient in the 40-clip environment.

C. DENSE Dataset: Algorithm Comparison at 1,000 Episodes

Tables V and VI present DENSE results at fixed hyperparameters (α=0.1, γ=0.6, ϵ=0.1) over 1,000 training episodes.

TABLE V: DENSE results with reward shaping, 1,000 episodes.

Algorithm	CR_{20}	CR_{30}	**Err. Cl.6**	**SD**
Q-Learning	17/39 (44%)	27/39 (69%)	23.7°	±14.3°
SARSA	17/39 (44%)	21/39 (54%)	24.0°	±15.5°
Double Q-Lrn.	16/39 (41%)	23/39 (59%)	25.1°	±15.6°
Paper [8]	N/A	N/A	57.0°	±2.8°

TABLE VI: DENSE results without reward shaping, 1,000 episodes.

Algorithm	CR_{20}	CR_{30}	**Err. Cl.6**	**SD**
Q-Learning	10/39 (26%)	16/39 (41%)	31.7°	±15.6°
SARSA	*25/39 (64%)*	*29/39 (74%)*	*18.5*°	±*12.1*°
Double Q-Lrn.	18/39 (46%)	24/39 (62%)	22.8°	±15.6°
Paper [8]	N/A	N/A	57.0°	±2.8°

All configurations beat the paper baseline of 57.0 degrees. The standout feature is the SARSA with no shaping configuration at 18.5 degrees (25 out of 39 subjects within 20 degrees), which demonstrates a distinct contrast with the SARSA with shaping configuration, which operates at 24.0 degrees (17 out of 39). This stands in contrast to DEAP, where reward shaping offers the maximum boost to the Double Q-Learning algorithm. In addition, reward shaping boosts the Q-Learning configuration on DENSE from 31.7 degrees to 23.7 degrees; yet, even with this improvement, the boosted Q-Learning configuration cannot match the best SARSA configuration.

D. Per-Subject Trajectory Analysis on DENSE

Table VII shows per-subject Q-Learning results for DENSE. The main sequence observed is anger $\rightarrow$ fear $\rightarrow$ pleasant

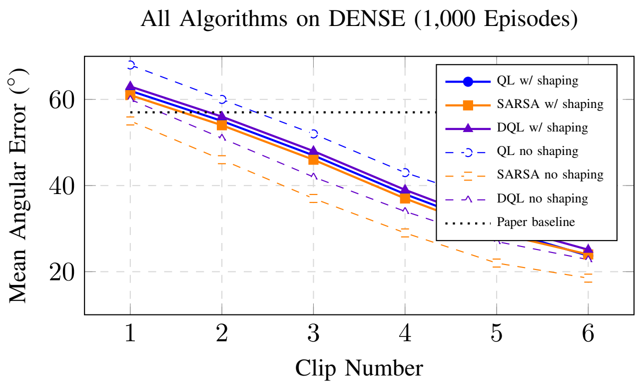

Fig. 4: Angular error clip by clip on DENSE (1,000 episodes). The error is lowest for unshaped SARSA at an average of 18.5 degrees (orange dashed line). All cases outperform the baseline provided in the paper by at least 57 degrees.

→ happy. The subjects who show a significantly lower error value ($< 10°$) within four clips are S13, S16, S23, S21, S05, and S09. It indicates that the transition model learned through LOSO for each individual fits their actual EEG-based reaction very well.

TABLE VII: Per-subject results: Q-Learning with shaping on DENSE, 1,000 episodes. Subjects are marked as within 20° or within 30°.

S.	Err	<20	<30	S.	Err	<20	<30
S01	66.3	–	–	S21	1.6	✓	✓
S02	39.8	–	–	S22	6.4	✓	✓
S03	29.4	–	✓	S23	3.8	✓	✓
S04	40.6	–	–	S24	24.3	–	✓
S05	9.4	✓	✓	S25	20.8	–	✓
S06	20.8	–	✓	S26	17.8	✓	✓
S07	23.3	–	✓	S27	33.0	–	–
S08	21.7	–	✓	S28	25.2	–	✓
S09	16.1	✓	✓	S29	14.4	✓	✓
S10	25.2	–	✓	S30	37.2	–	–
S11	40.3	–	–	S31	25.2	–	✓
S12	37.2	–	–	S32	34.8	–	–
S13	5.1	✓	✓	S33	5.8	✓	✓
S14	18.5	✓	✓	S34	17.9	✓	✓
S15	11.0	✓	✓	S35	10.3	✓	✓
S16	7.3	✓	✓	S36	34.2	–	–
S17	43.3	–	–	S37	42.5	–	–
S18	25.2	–	✓	S38	15.2	✓	✓
S19	16.3	✓	✓	S39	48.6	–	–
S20	10.0	✓	✓				
Summary: 17/39 within $20°$**, 27/39 within** $30°$**, Mean** $23.7° \pm 14.3°$							

E. DENSE: SARSA Per-Subject Results (Best Condition)

In Table VIII, the breakdown results for SARSA without shaping have been provided, pinpointing the highest-performing setting amongst all subjects in DENSE. The dominant pathway continues to be anger → fear → pleasant → happy. As compared to Q-Learning, due to the on-policy nature of SARSA, there are significantly fewer fluctuations in performance. This is reflected in the fact that out of a total of 39 subjects, 25 fall within 20° from the optimal solution, whereas the standard deviation (+-12.1°) is significantly lower than +15.6° observed for Q-Learning without shaping.

TABLE VIII: Per-subject results: SARSA without shaping on DENSE, 1,000 episodes. This is the best overall configuration for DENSE.

S.	Err	<20	<30	S.	Err	<20	<30
S01	30.7	–	–	S21	11.7	✓	✓
S02	9.5	✓	✓	S22	38.3	–	–
S03	38.3	–	–	S23	6.6	✓	✓
S04	2.7	✓	✓	S24	13.7	✓	✓
S05	0.1	✓	✓	S25	21.7	–	✓
S06	8.1	✓	✓	S26	22.7	–	✓
S07	16.5	✓	✓	S27	38.3	–	–
S08	3.4	✓	✓	S28	38.3	–	–
S09	7.3	✓	✓	S29	11.6	✓	✓
S10	18.1	✓	✓	S30	16.0	✓	✓
S11	15.3	✓	✓	S31	38.3	–	–
S12	42.0	–	–	S32	22.3	–	✓
S13	3.9	✓	✓	S33	32.6	–	–
S14	12.7	✓	✓	S34	11.1	✓	✓
S15	13.2	✓	✓	S35	15.3	✓	✓
S16	31.4	–	–	S36	18.3	✓	✓
S17	13.2	✓	✓	S37	14.8	✓	✓
S18	38.3	–	–	S38	22.6	–	✓
S19	17.9	✓	✓	S39	2.2	✓	✓
S20	4.2	✓	✓				
Summary: 25/39 within $20°$**, 29/39 within** $30°$**, Mean** $18.5° \pm 12.1°$							

F. Master Comparison: All Conditions

Table IX captures all key experimental parameters. In either data set, no method is uniformly superior: Double Q-Learning excels on the controlled DEAP environment, but SARSA outperforms when no shaping is used on the difficult DENSE environment.

TABLE IX: Complete experimental summary across datasets, algorithms, and shaping conditions. QL=Q-Learning, DQL=Double Q-Learning. Best per dataset shown in italics.

Data	Algo	Sh.	Ep.	CR_{20}	CR_{30}	Cl.6 Err.	vs Paper
DEAP	QL	Yes	3k	21/32	23/32	20.1°±23.5°	−65%
DEAP	QL	No	3k	24/32	26/32	13.9°±18.2°	−76%
DEAP	SARSA	Yes	3k	29/32	29/32	13.8°±26.8°	−76%
DEAP	SARSA	No	3k	27/32	28/32	21.5°±22.4°	−62%
DEAP	*DQL*	*Yes*	*3k*	*30/32*	*32/32*	*8.3°±7.4°*	*−85%*
DEAP	DQL	No	3k	17/32	21/32	49.2°±38.5°	−14%
DENSE	QL	Yes	1k	17/39	27/39	23.7°±14.3°	−58%
DENSE	QL	No	1k	10/39	16/39	31.7°±15.6°	−44%
DENSE	SARSA	Yes	1k	17/39	21/39	24.0°±15.5°	−58%
DENSE	*SARSA*	*No*	*1k*	*25/39*	*29/39*	*18.5°±12.1°*	*−68%*
DENSE	DQL	Yes	1k	16/39	23/39	25.1°±15.6°	−56%
DENSE	DQL	No	1k	18/39	24/39	22.8°±15.6°	−60%
Paper [8] (DEAP, QL, w/ shaping)				19/32	N/A	57.0°±2.8°	–

VII. Discussion

A. Why Double Q-Learning Excels on DEAP

The key factor responsible for the success of Double Q-Learning in DEAP environments is the relation between the size of the action space and noise associated with estimating Q-values. In an environment where 40 music clips are involved, the learning space is high-dimensional and the estimation errors that are associated with Q-values are correlated when there is only one Q-table at the initial stages of learning. By employing two Q-tables in Double Q-Learning, this problem can be addressed because the noise realization will be different from each other when choosing actions using Q1 and evaluating values using Q2.

It is interesting to note that this particular benefit disappears without the presence of the shaping signal. The removal of the penalty for stagnation leads to DEAP results of Double Q-Learning deteriorating from 8.3° to 49.2°, which is actually lower than that obtained using an unoptimized version of Q-Learning. Although the dual tables do well at filtering out noise, it seems that they need the guidance that the shaping gradient gives to achieve meaningful improvements during training in sparse reward environments.

B. Why SARSA Leads on DENSE

SARSA shows itself superior to DENSE traces based on one particular feature of on-policy learning, which proves to be particularly beneficial when dealing with heterogenous data. As compared to DEAP, the DENSE generalization problem is harder: held-out subjects always start with Anger; in addition, the transition dynamics of the EEGs appear more specific due to the inherent variety of the real-world scenario. Here, using a Q-learning algorithm to always bootstrap towards optimistic Q-values can lead to the overvaluation of Q-estimates, as they may fail to generalize onto the held-out subject. On the contrary, the SARSA update formula takes into account the price paid for non-optimal exploration, resulting in Q-estimates that are not only more pessimistic but also generalizable to other subjects from the LOSO training set. It must be noted that the lower standard deviation obtained by SARSA (±12.1°) compared to that of Q-Learning without shaping (±15.6°) should not go unnoticed. From a practical application standpoint, the variability in outcomes is almost as important as the average value itself; an algorithm that occasionally misleads subjects into the wrong affective quadrant can be distinguished from an algorithm that successfully converges for most patients even if both algorithms have similar average results. Adding reward shaping effectively ties SARSA with Q-Learning on DENSE ($24.0°$ vs. $23.7°$), suggesting that the stagnation potential partially compensates for Q-Learning's overestimation in the smaller 7-clip action space, but at the cost of eroding SARSA's distributional advantage. This is consistent with the findings of Grzes and Kudenko [20] that imperfectly specified shaping potentials can interfere with on-policy convergence. The angular-distance potential $\Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2$ is geometrically well-motivated but not precisely calibrated to the discrete DENSE transition graph, so SARSA's direct on-policy experience turns out to be a more reliable guide than the potential function in this case.

C. The Reward Shaping Interaction

A clear trend appears to exist in all six algorithm-dataset combinations; shaping offers the largest benefit when applied to the algorithm suffering the worst case of overestimation. When applied to Q-Learning (off-policy, single-table), there exists a substantial benefit in both datasets. Shaping offers a large benefit to Double Q-Learning on the DEAP dataset, since the wide action space increases overestimation, but the benefit decreases on the DENSE dataset, where the narrow action space limits the errors that an off-policy algorithm can make. The benefit to SARSA, on the other hand, which uses an on-policy algorithm and therefore avoids overestimation, is minimal.

D. Algorithm Selection Guidelines

From these experiments, several practical recommendations emerge for practitioners building EEG-driven music therapy systems:

1) **Large action spaces (40+ clips): use Double Q-Learning with reward shaping.** Dual Table Variance Reduction combined with Stagnation Penalty results in the fastest rate of convergence with the narrowest spread of results. In the DEAP algorithm, this combination achieved CR30 = 100% for 0.10.
2) **Smaller or naturalistic datasets (7 emotion categories, fewer clips): use SARSA without shaping.** On-policy conservatism outperforms both Q-Learning and Double Q-Learning when subject heterogeneity is the primary challenge, particularly when the episode budget is constrained.
3) **Never set $\epsilon = 0$.** Exploration was completely unsuccessful in all of the algorithms tested and across all hyperparameters used. In this case, exploration cannot be treated as something to fine-tune; instead, it must be done. The parameters between 0.10 and 0.20 always provided good results.
4) **Plan for at least 1,000 training episodes per LOSO fold.** Performance at 300 episodes shows significantly lower efficiency compared to the increasing trend seen between 1,000 and 3,000 episodes. Datasets with training subjects less than 25 require more episodes.
5) **Consider a neutral warm-up phase for Anger-starting subjects.** With respect to DENSE, the state Anger presents the hardest starting state. By offering a few neutral emotions before the start of the RL-based recommendation process, we can place the individuals in a more manageable area on the V-A graph, thus requiring fewer recommendations.

VIII. Limitations and Future Work

There are several factors that limit the validity of the findings reported here. For instance, both the databases divide the emotions in the DEAP database into a 20×20 matrix, and in DENSE into seven clusters, losing the finer granularity in the continuous V-A spectrum that would allow differentiation between similar emotions such as contentment and mild happiness. Another constraint on DENSE is the fixed initial state of Anger for all participants, making it unclear whether the results can be generalized to different initial states.

Deployment readiness is also impacted by the data collection mode. Both the DEAP and DENSE datasets use a controlled environment for data acquisition, together with prescribed annotation frameworks; neither provides an actual real-time assessment within a natural setting at home or within a clinical audiometry environment. What is needed to complete the deployment cycle is the implementation of a real-time EEG classification module. The BiLSTM network trained on the DEAP dataset has attained a test accuracy rate of around 94%; thus, it appears feasible to implement this within the recommender framework.

Future avenues include live-stream emotion classification through EEG signals suitable for implementation in a web browser; using Deep Q-Networks to control the continuous V-A space without segmentation; expanding the clip database to cover hundreds of clips spanning a wider range of V-A distributions; subject-specific potential for learning determined during an enrolment session; and conducting randomized controlled studies in clinical groups to assess the treatment effects compared to current standards of care. Wearable consumer EEG equipment used together with streamlined inference would be a logical direction to go.

IX. Conclusion

In order to compare the performance of the Q-learning, SARSA, and Double Q-learning reinforcement learning techniques on the problem of regulating emotions with music using EEG, a detailed study was carried out involving two different datasets having different structural characteristics.

In the case of DEAP experiment with 3,000 episodes, Double Q-Learning with reward shaping produced the angular error of 8.3 degrees with standard deviation of 7.4 degrees, resulting in convergence for 30 out of 32 participants up to 20 degrees and for all 32 participants up to 30 degrees, which is an 85% improvement compared to the baseline from Dutta et al. (2020). In the more difficult experiment on the DENSE dataset with 1,000 episodes, SARSA without reward shaping obtained angular error of 18.5 degrees with standard deviation of 12.1 degrees, producing convergence for 25 out of 39 participants in 20 degrees, achieving 68% performance gain despite using the naturalistic database.

The main finding is that the agreement between algorithm and data plays an important role for their effectiveness, since there is no one-size-fits-all solution. The design of Double Q-Learning with two lookup tables appears to be most advantageous when overestimation is the key problem, like when there are large action spaces and small amounts of data available. The on-policy approach used by SARSA exhibits more robustness in the cases where heterogeneity of subjects and noisy state transitions are the key challenges.

Combined, these results suggest that appropriately constructed tabular reinforcement learning (RL) could be sufficient to reliably guide emotional states towards positively valued goals, to an extent that should be taken seriously in a clinical setting. It follows logically to combine the use of a real-time EEG-based emotion classification system with the best RL constructions identified through the course of this research.

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